

Determinant-maximizing frames and finite recovery

[Back to Project mathematical routes](#)

Scope

An attained positive determinant maximum supplies well-conditioned dual frames. Weighted satellite arguments and a finite almost-norming family control the vectors needed for approximate linear recovery; the exact lower-unit condition remains explicit.

11 direct Cards + 1 reused prerequisite = 12 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [A finite absolute-maximizing family almost norms the unit sphere](#)
- Key result: [A common inverse estimate from row Cramer](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#) [4](#)

Level 0 (3 Cards)

Level 0

A weighted determinant polynomial with satellites

Adds linear row-replacement contributions to a base determinant.

`MathlibAnnex.Satellite.satellitePolynomial`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Absolute maximization forces a near-maximal base](#) · [Each satellite norms its coefficient preimage in absolute value](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

An attained absolute determinant maximum

Selects a maximizing dual frame for a specified basis.

`MathlibAnnex.DeterminantFrame.exists_maximizingFrame`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Absolute maximization forces a near-maximal base](#) · [The set of near-maximal dual frames](#) · [Positivity of the determinant maximum](#)

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Level 0

A seminorm with two-sided bounds against a reference norm

Records a norm through a seminorm, two positive comparison constants and continuity on the original normed space.

`MathlibAnnex.EquivalentSeminorm`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A strict lower bound on a norm's unit sphere](#) · [Absolute maximization forces a near-maximal base](#) · [Each satellite norms its coefficient preimage in absolute value](#) · [A nearby norming point gives an almost-norming evaluation](#)

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Level 1 (5 Cards)

Level 1

Absolute maximization forces a near-maximal base

A dominant determinant term prevents a large deficit in the base frame.

`MathlibAnnex.Satellite.absoluteMaximizer_base_nearMax`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#) · [A weighted determinant polynomial with satellites](#) · [An attained absolute determinant maximum](#)

Used by in this scope: [A finite absolute-maximizing family almost norms the unit sphere](#)

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Level 1

Positivity of the determinant maximum

A uniformly scaled dual basis supplies a positive comparison determinant.

`MathlibAnnex.DeterminantFrame.determinantMaximum_pos`

Immediate Card prerequisites: [An attained absolute determinant maximum](#)

Used by in this scope: [A common inverse estimate from row Cramer](#)

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Level 1

A strict lower bound on a norm's unit sphere

Defines the strict lower condition on a fixed norm unit sphere.

`MathlibAnnex.FiniteSup.FinMapAlmostIsometric`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 1

The set of near-maximal dual frames

Records a determinant deficit while keeping all dual rows contractive.

`MathlibAnnex.DeterminantFrame.nearMaxFrames`

Immediate Card prerequisites: [An attained absolute determinant maximum](#)

Used by in this scope: [A common inverse estimate from row Cramer](#) · [Each satellite norms its coefficient preimage in absolute value](#)

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Level 1

A nearby norming point gives an almost-norming evaluation

Two triangle inequalities transfer absolute norm attainment to a unit vector.

`MathlibAnnex.Satellite.abs_eval_lower_of_norms_nearby`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [One fixed satellite family almost norms every detected unit vector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2 (2 Cards)

Level 2

A common inverse estimate from row Cramer

Controls all near-maximal inverse frames by an explicit basis-dependent constant.

`MathlibAnnex.DeterminantFrame.inverseBoundConstant_bound`

Immediate Card prerequisites: [The set of near-maximal dual frames](#) · [Positivity of the determinant maximum](#)

Used by in this scope: [A finite absolute-maximizing family almost norms the unit sphere](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Each satellite norms its coefficient preimage in absolute value

Varying one functional at an absolute maximizer forces endpoint attainment.

`MathlibAnnex.Satellite.absoluteMaximizer_satellite_norms_preimage`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#) · [A weighted determinant polynomial with satellites](#) · [The set of near-maximal dual frames](#)

Used by in this scope: [One fixed satellite family almost norms every detected unit vector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (1 Card)

Level 3

One fixed satellite family almost norms every detected unit vector

Uses a detecting coefficient and its matching satellite without replacing the family.

```
MathlibAnnex.Satellite.absoluteMaximizer_satellite  
s_almost_norm
```

Immediate Card prerequisites: [Each satellite norms its coefficient preimage in absolute value](#) · [A nearby norming point gives an almost-norming evaluation](#)

Used by in this scope: [A finite absolute-maximizing family almost norms the unit sphere](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4 (1 Card)

Level 4

A finite absolute-maximizing family almost norms the unit sphere

Chooses the inverse bound, net and weight before fixing one satellite family.

```
MathlibAnnex.Satellite.exists_goodSatellitePackage
```

Immediate Card prerequisites: [A common inverse estimate from row Cramer](#) · [Absolute maximization forces a near-maximal base](#) · [One fixed satellite family almost norms every detected unit vector](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)